数据分析
Data Analytics

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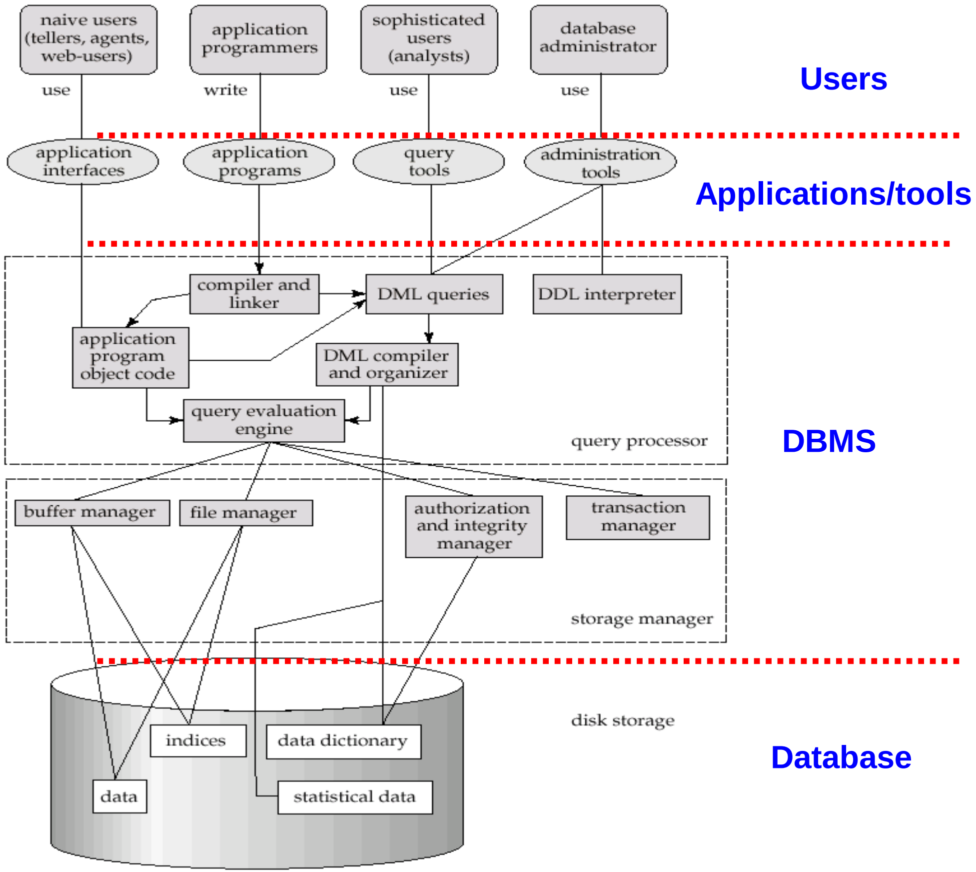
课程概要

- **Part 0: Overview**
  - Ch1: Introduction
- **Part 1 Relational Languages**
  - Ch2: Relational model
  - Ch3: Introduction to SQL
  - Ch4: Intermediate SQL
  - Ch5: Advanced SQL
- **Part 2 Database Design**
  - Ch6: Database design via E-R model
  - Ch7: Relational database design
- **Part 3 Application Design & Development**
  - Ch8: Complex data types
  - Ch9: Application development
- **Part 4 Big Data Analytics**
  - Ch10: Big data
  - **Ch11: Data analytics**
- **Part 5 Storage Management & Indexing**
  - Ch12: Physical storage systems
  - Ch13: Data storage structures
  - Ch14: Indexing
- **Part 6 Query Processing & Optimization**
  - Ch15: Query processing
  - Ch16: Query optimization
- **Part 7 Transaction Management**
  - Ch17: Transactions
  - Ch18: Concurrency control
  - Ch19: Recovery system
- **Part 8 Parallel & Distributed Database**
  - Ch20: Database system architecture
  - Ch21-23: Parallel & distributed storage, query processing & transaction processing
- **Advanced topics**
  - DB Platform: **OceanBase**, MongoDB, Neo4J
  - …

Database System Structure



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Overview

- **Data analytics**: the processing of data to infer patterns, correlations, or models for prediction
- Primarily used to make business decisions
  - Per individual customer
    - E.g., what product to suggest for purchase
  - Across all customers
    - E.g., what products to manufacture/stock, in what quantity
- Critical for businesses today

Overview (Cont.)

- **Common steps in data analytics**
  - Gather data from multiple sources into one location
    - Data warehouses also integrated data into common schema
    - Data often needs to be **extracted** from source formats, **transformed** to common schema, and **loaded** into the data warehouse
      - Can be done as **ETL (extract-transform-load)**, or **ELT (extract-load-transform)**
  - Generate aggregates and reports summarizing data
    - Dashboards showing graphical charts/reports
    - **Online analytical processing (OLAP) systems** allow interactive querying
    - Statistical analysis using tools such as R/SAS/SPSS
      - Including extensions for parallel processing of big data
  - Build **predictive models** and use the models for decision making

Overview (Cont.)

- Predictive models are widely used today
  - E.g., use customer profile features (e.g. income, age, gender, education, employment) and past history of a customer to predict likelihood of default on loan
    - and use prediction to make loan decision
  - E.g., use past history of sales (by season) to predict future sales
    - And use it to decide what/how much to produce/stock
    - And to target customers
- Other examples of business decisions:
  - What items to stock?
  - What insurance premium to change?
  - To whom to send advertisements?

Overview (Cont.)

- **Machine learning** techniques are key to finding patterns in data and making predictions
- **Data mining** extends techniques developed by machine-learning communities to run them on very large datasets
- The term **business intelligence (BI)** is synonym for data analytics
- The term **decision support** focuses on reporting and aggregation

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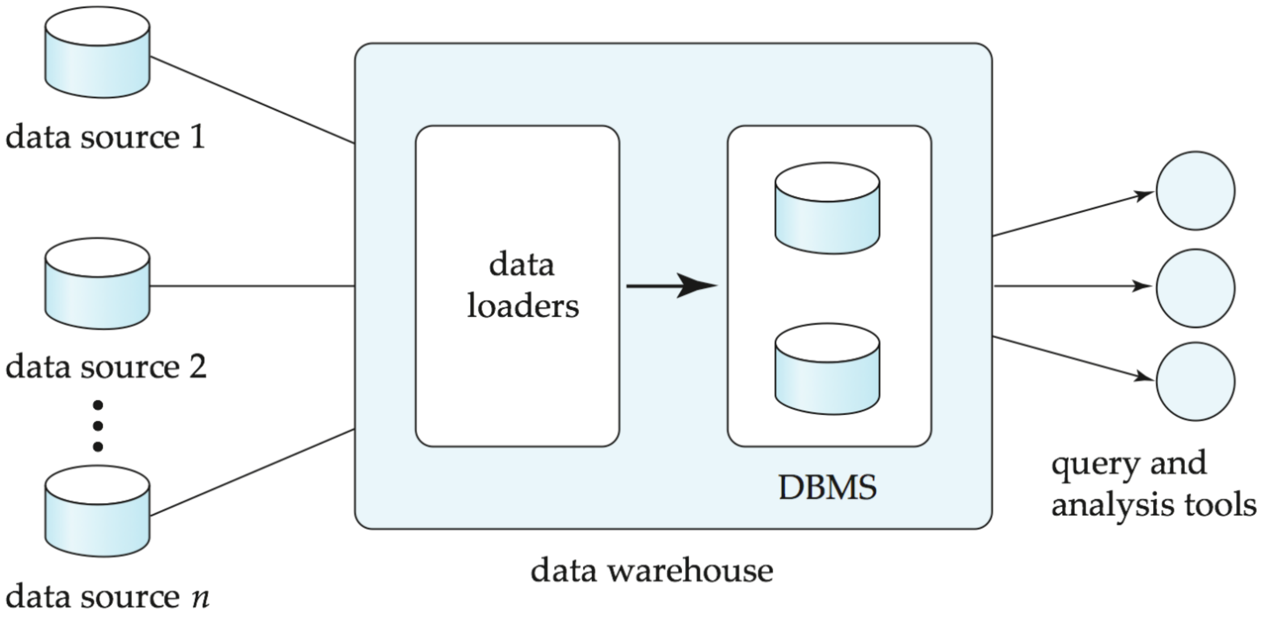
- 数据仓库
- 联机分析处理
- 数据挖掘

Data Warehousing

- Data sources often store only current data, not historical data
- Corporate decision making requires a unified view of all organizational data, including historical data
- A **data warehouse** is a repository (archive) of information gathered from multiple sources, stored under a unified schema, at a single site
  - Greatly simplifies querying, permits study of historical trends
  - Shifts decision support query load away from transaction processing systems

Data Warehousing



Design Issues

- *When and how to gather data*
  - **Source driven architecture**: data sources transmit new information to warehouse
    - either continuously or periodically (e.g., at night)
  - **Destination driven architecture:** warehouse periodically requests new information from data sources
  - **Synchronous** vs **asynchronous replication**
    - Keeping warehouse exactly synchronized with data sources (e.g., using two-phase commit) is often too expensive
    - Usually OK to have slightly out-of-date data at warehouse
    - Data/updates are periodically downloaded form online transaction processing (OLTP) systems.
- *What schema to use*
  - Schema integration

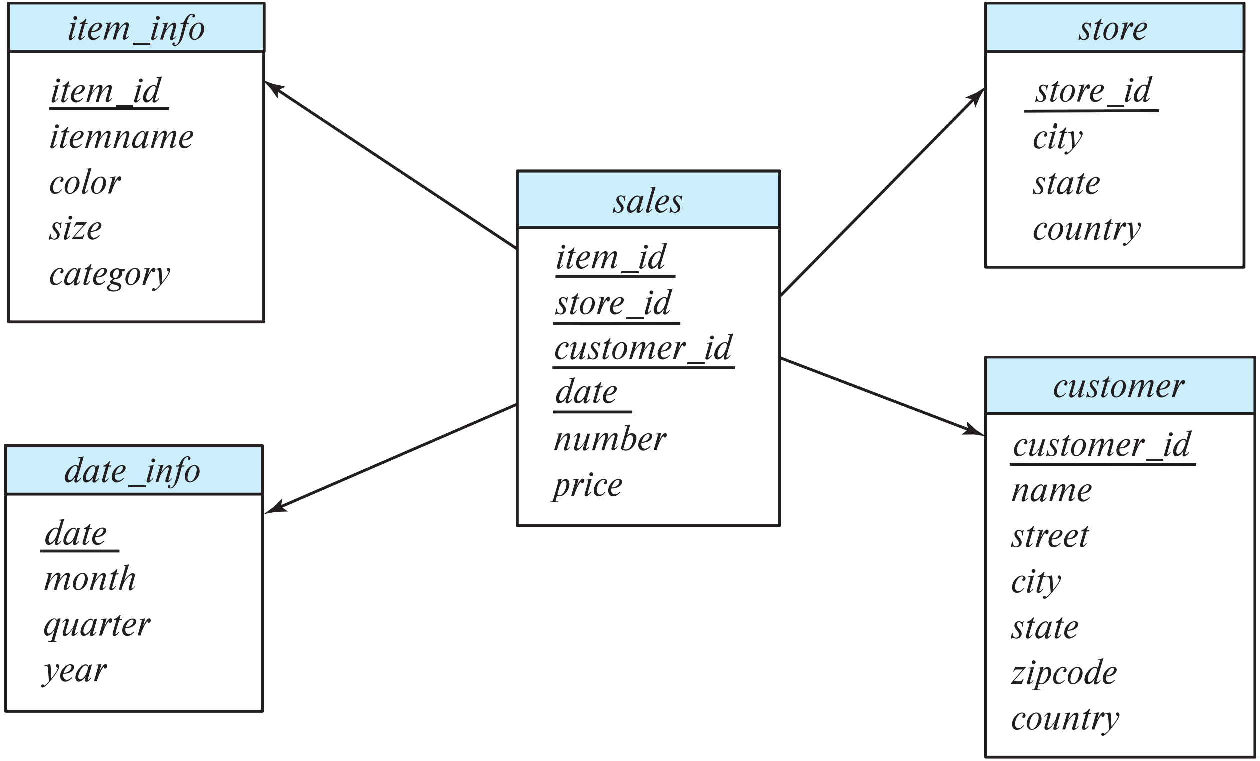
More Warehouse Design Issues

- **Data transformation** and **data cleansing**
  - E.g., correct mistakes in addresses (misspellings, zip code errors)
  - Merge address lists from different sources and purge duplicates
- How to propagate updates
  - Warehouse schema may be a (materialized) view of schema from data sources
    - View maintenance
- *What data to summarize*
  - Raw data may be too large to store on-line
  - Aggregate values (totals/subtotals) often suffice
  - Queries on raw data can often be transformed by query optimizer to use aggregate values

Multidimensional Data and Warehouse Schemas

- Data in warehouses can usually be divided into
  - **Fact tables**, which are large
    - E.g, *sales*(*item_id, store_id, customer_id, date, number, price*)
  - **Dimension tables**, which are relatively small
    - Store extra information about stores, items, etc.
- Attributes of fact tables can be usually viewed as
  - **Measure attributes**
    - measure some value, and can be aggregated upon
    - e.g., the attributes *number* or *price* of the *sales* relation
  - **Dimension attributes**
    - dimensions on which measure attributes are viewed
    - e.g., attributes *item_id, color,* and *size* of the *sales* relation
    - Usually small ids that are foreign keys to dimension tables

Data Warehouse Schema



Multidimensional Data and Warehouse Schemas

- Resultant schema is called a **star schema**
  - More complicated schema structures
    - **Snowflake schema**: multiple levels of dimension tables
    - May have multiple fact tables
- Typically
  - fact table joined with dimension tables and then
  - group-by on dimension table attributes, and then
  - aggregation on measure attributes of fact table
- Some applications do not find it worthwhile to bring data to a common schema
  - **Data lakes** are repositories which allow data to be stored in multiple formats, without schema integration
  - Less upfront effort, but more effort during querying

Database Support for Data Warehouses

- Data in warehouses usually append only, not updated
  - Can avoid concurrency control overheads
- Data warehouses often use **column-oriented storage**
  - E.g., a sequence of *sales* tuples is stored as follows
    - Values of item_id attribute are stored as an array
    - Values of store_id attribute are stored as an array,
    - And so on
  - Arrays are compressed, reducing storage, IO and memory costs significantly
  - Queries can fetch only attributes that they care about, reducing IO and memory cost
  - More details in Section 13.6
- Data warehouses often use parallel storage and query processing infrastructure
  - Distributed file systems, Map-Reduce, Hive, …

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Data Analysis and OLAP

- **Online Analytical Processing (OLAP)**
  - Interactive analysis of data, allowing data to be summarized and viewed in different ways in an online fashion (with negligible delay)
- We use the following relation to illustrate OLAP concepts
  - *sales* (*item_name*, *color*, *clothes_size*, *quantity*)

  This is a simplified version of the *sales* fact table joined with the dimension tables, and many attributes removed (and some renamed)

Example sales relation

item_name	*color*	*clothes_size*	*quantity*
dress	dark	small	2
dress	dark	medium	6
dress	dark	large	12
dress	pastel	small	4
dress	pastel	medium	3
dress	pastel	large	3
dress	white	small	2
dress	white	medium	3
dress	white	large	0
pants	dark	small	14
pants	dark	medium	6
pants	dark	large	0
pants	pastel	small	1
pants	pastel	medium	0
pants	pastel	large	1
pants	white	small	3
pants	white	medium	0
pants	white	large	2
shirt	dark	small	2
shirt	dark	medium	6
shirt	dark	large	6
shirt	pastel	small	4
shirt	pastel	medium	1
shirt	pastel	large	2
shirt	white	small	17
shirt	white	medium	1
shirt	white	large	10
skirt	dark	small	2
skirt	dark	medium	5

Cross Tabulation of *sales* by *item_name* and *color*

clothes_size **all**

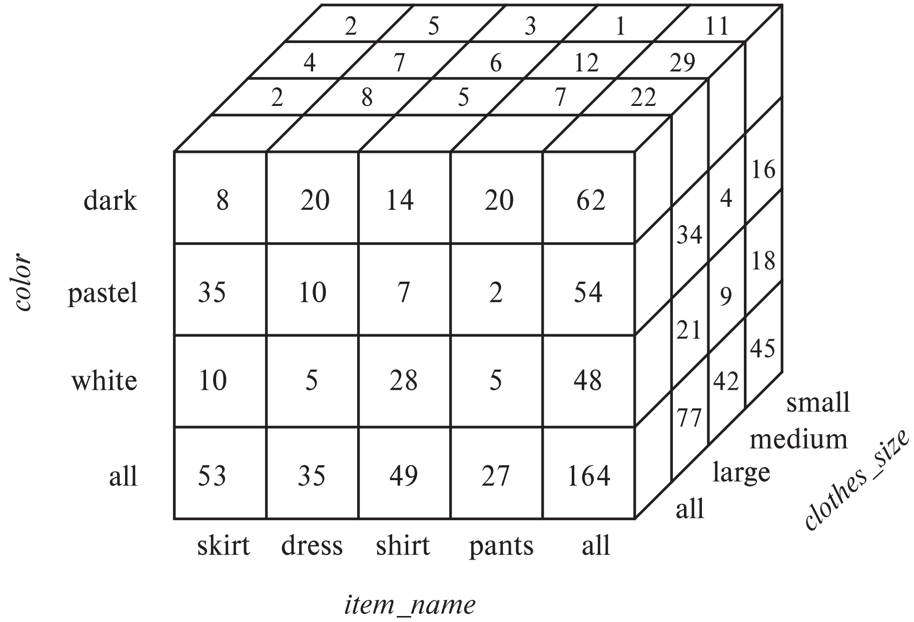
color

item_name	dark	pastel	white	total
skirt	8	35	10	53
dress	20	10	5	35
shirt	14	7	28	49
pants	20	2	5	27
total	62	54	48	164

- The table above is an example of a **cross-tabulation** (**cross-tab**), also referred to as a **pivot-table.**
  - Values for one of the dimension attributes form the row headers
  - Values for another dimension attribute form the column headers
  - Other dimension attributes are listed on top
  - Values in individual cells are (aggregates of) the values of the dimension attributes that specify the cell.

Data Cube

- A **data cube** is a multidimensional generalization of a cross-tab
- Can have n dimensions; we show 3 below
- Cross-tabs can be used as views on a data cube



Online Analytical Processing Operations

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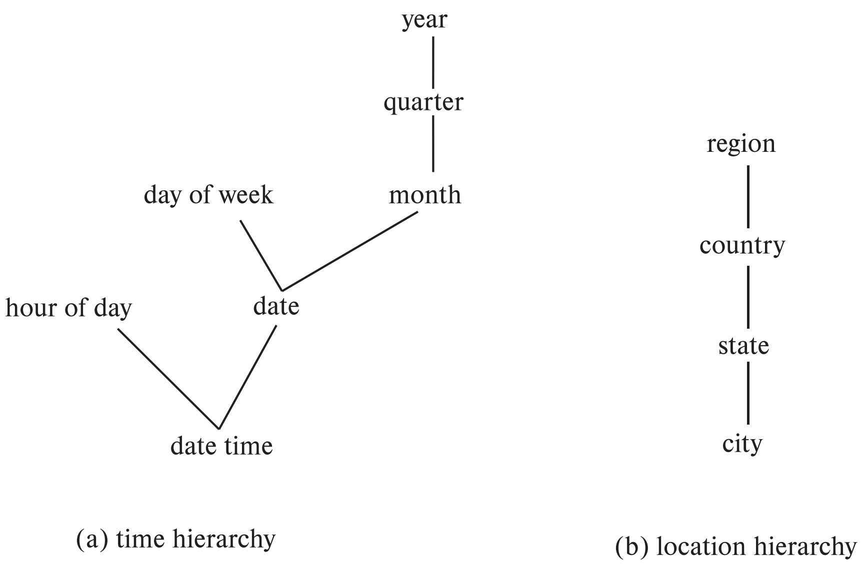
- **Pivoting:** changing the dimensions used in a cross-tab
  - E.g., moving colors to column names
- **Slicing:** creating a cross-tab for fixed values only
  - E.g., fixing color to white and size to small
  - Sometimes called **dicing**, particularly when values for multiple dimensions are fixed.

Online Analytical Processing Operations

- **Rollup:** moving from finer-granularity data to a coarser granularity
  - E.g., aggregating away an attribute
  - E.g., moving from aggregates by day to aggregates by month or year
- **Drill down:** The opposite operation - that of moving from coarser-granularity data to finer-granularity data

Hierarchies on Dimensions

- **Hierarchy** on dimension attributes: lets dimensions be viewed at different levels of detail
- E.g., the dimension *datetime* can be used to aggregate by hour of day, date, day of week, month, quarter or year



Cross Tabulation With Hierarchy

- Cross-tabs can be easily extended to deal with hierarchies
- Can drill down or roll up on a hierarchy
- E.g. hierarchy: *item_name* → *category*

clothes_size: **all**

category	*item_name*	*color* dark	pastel	white	total	
womenswear	skirt	8	8	10	53	
	dress	20	20	5	35	
	subtotal	28	28	15		88
menswear	pants	14	14	28	49	
	shirt	20	20	5	27	
	subtotal	34	34	33		76
total		62	62	48		164

Relational Representation of Cross-tabs

- Cross-tabs can be represented as relations
- We use the value **all** to represent aggregates.
- The SQL standard actually uses *null* values in place of **all**
  - Works with any data type
  - But can cause confusion with regular null values.

item_name	*color*	*clothes_size*	*quantity*
skirt	dark	**all**	8
skirt	pastel	**all**	35
skirt	white	**all**	10
skirt	**all**	**all**	53
dress	dark	**all**	20
dress	pastel	**all**	10
dress	white	**all**	5
dress	**all**	**all**	35
shirt	dark	**all**	14
shirt	pastel	**all**	7
shirt	white	**all**	28
shirt	**all**	**all**	49
pants	dark	**all**	20
pants	pastel	**all**	2
pants	white	**all**	5
pants	**all**	**all**	27
all	dark	**all**	62
all	pastel	**all**	54
all	white	**all**	48
all	**all**	**all**	164

Pivot Operation

- **select** *
  from *sales*
  pivot (
      sum(*quantity*)
      for *color* **in** ('dark','pastel','white')
)
  order by *item name*;

item_name	*clothes_size*	*dark*	*pastel*	*white*
dress	small	2	4	2
dress	medium	6	3	3
dress	large	12	3	0
pants	small	14	1	3
pants	medium	6	0	0
pants	large	0	1	2
shirt	small	2	4	17
shirt	medium	6	1	1
shirt	large	6	2	10
skirt	small	2	11	2
skirt	medium	5	9	5
skirt	large	1	15	3

Cube Operation

- The **cube** operation computes union of **group by**'s on every subset of the specified attributes
- E.g., consider the query

  select *item_name, color, size,* **sum**(*number*)
  from *sales*
  group by cube(*item_name, color, size*)

This computes the union of eight different groupings of the *sales* relation:

{ (*item_name, color, size*), (*item_name, color*),
(*item_name, size*), (*color, size*),
(*item_name*), (*color*),
(*size*), ( ) }

where ( ) denotes an empty **group by** list.

- For each grouping, the result contains the null value for attributes not present in the grouping

Online Analytical Processing Operations

- Relational representation of cross-tab that we saw earlier, but with *null* in place of **all**, can be computed by

  select *item_name*, *color*, **sum**(*number*)
  from *sales*
  group by cube(*item_name, color*)

- The function **grouping()** can be applied on an attribute
  - Returns 1 if the value is a null value representing all, and returns 0 in all other cases.

  select case when grouping(*item_name*) = 1 **then** '**all**'
  else *item_name* **end as** *item_name*,
  case when grouping(*color*) = 1 **then** '**all**'
  else *color* **end as** *color*,
  '**all**' **as** *clothes size*, **sum**(*quantity*) **as** *quantity*
  from *sales*
  group by cube(*item name*, *color*);

Online Analytical Processing Operations

- Can use the function **decode()** in the **select** clause to replace such nulls by
  a value such as **all**
  - E.g., replace *item_name* in first query by

    decode(**grouping**(item_*name*), 1, 'all', *item_name*)

Extended Aggregation (Cont.)

- The **rollup** construct generates union on every prefix of specified list of attributes

  select *item_name*, *color*, *size*, **sum**(*number*)
  from *sales*
  group by rollup(*item_name, color, size*)

  Generates union of four groupings:

  { (*item_name, color, size*), (*item_name, color*), (*item_name*), ( ) }
- Rollup can be used to generate aggregates at multiple levels of a hierarchy.
- E.g., suppose table *itemcategory*(*item_name, category*) gives the category of each item. Then

  select *category, item_name*, **sum**(*number*)
  from *sales, itemcategory*
  where *sales.item_name = itemcategory.item_name*
  group by rollup(*category, item_name*)

  would give a hierarchical summary by *item_name* and by *category.*

Extended Aggregation (Cont.)

- Multiple rollups and cubes can be used in a single group by clause
  - Each generates set of group by lists, cross product of sets gives overall set of group by lists
- E.g.,

select *item_name, color, size*, **sum**(*number*)
from *sales*
group by rollup(*item_name*), **rollup**(*color, size*)

generates the groupings

{*item_name, ()*} *X* {*(color, size), (color), ()*}

= { (*item_name, color, size*), (*item_name, color*), (*item_name*), (*color, size*), (*color*), ( ) }

select *item_name*, *color*, *clothes_size*, **sum**(*quantity*)
from *sales*
group by grouping sets ((*color*, *clothes_size*), (*clothes_size*, *item_name*));

OLAP Implementation

- The earliest OLAP systems used multidimensional arrays in memory to store data cubes, and are referred to as **multidimensional OLAP (MOLAP)** systems.
- OLAP implementations using only relational database features are called **relational OLAP (ROLAP)** systems
- Hybrid systems, which store some summaries in memory and store the base data and other summaries in a relational database, are called **hybrid OLAP (HOLAP)** systems.

OLAP Implementation (Cont.)

- Early OLAP systems precomputed *all* possible aggregates in order to provide online response
  - Space and time requirements for doing so can be very high
    - 2^n combinations of **group by**
  - It suffices to precompute some aggregates, and compute others on demand from one of the precomputed aggregates
    - Can compute aggregate on (*item_name, color*) from an aggregate on (*item_name, color, size*)
      - For all but a few "non-decomposable" aggregates such as *median*
      - is cheaper than computing it from scratch
- Several optimizations available for computing multiple aggregates
  - Can compute aggregate on (*item_name, color*) from an aggregate on (*item_name, color, size*)
  - Can compute aggregates on (*item_name, color, size*), (*item_name, color*) and (*item_name*) using a single sorting of the base data

Reporting and Visualization

- **Reporting tools** help create formatted reports with tabular/graphical representation of data
  - E.g., SQL Server reporting services, Crystal Reports
- **Data visualization** tools help create interactive visualization of data
  - E.g., Tableau, FusionChart, plotly, Datawrapper, Google Charts, etc.
  - Frontend typically based on HTML+JavaScript

Acme Supply Company, Inc.
Quarterly Sales Report

Period: Jan. 1 to March 31, 2009

Region	Category	Sales	Subtotal
North	Computer Hardware	1,000,000	
	Computer Software	500,000	
	All categories		1,500,000
South	Computer Hardware	200,000	
	Computer Software	400,000	
	All categories		600,000
		Total Sales	2,100,000

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Data Mining

- **Data mining** is the process of semi-automatically analyzing large databases to find useful patterns
  – Similar goals to machine learning, but on very large volumes of data
- Part of the larger area of **knowledge discovery in databases (KDD)**
- Some types of knowledge can be represented as rules
- More generally, knowledge is discovered by applying machine learning techniques on past instances of data, to form a **model**
  – Model is then used to make predictions for new instances

Types of Data Mining Tasks

- **Prediction** based on past history
  - Predict if a credit card applicant poses a good credit risk, based on some attributes (income, job type, age, ..) and past history
  - Predict if a pattern of phone calling card usage is likely to be fraudulent
- Some examples of prediction mechanisms:
  - **Classification**
    - Items (with associated attributes) belong to one of several classes
    - **Training instances** have attribute values and classes provided
    - Given a new item whose class is unknown, predict to which class it belongs based on its attribute values
  - **Regression** formulae
    - Given a set of mappings for an unknown function, predict the function result for a new parameter value

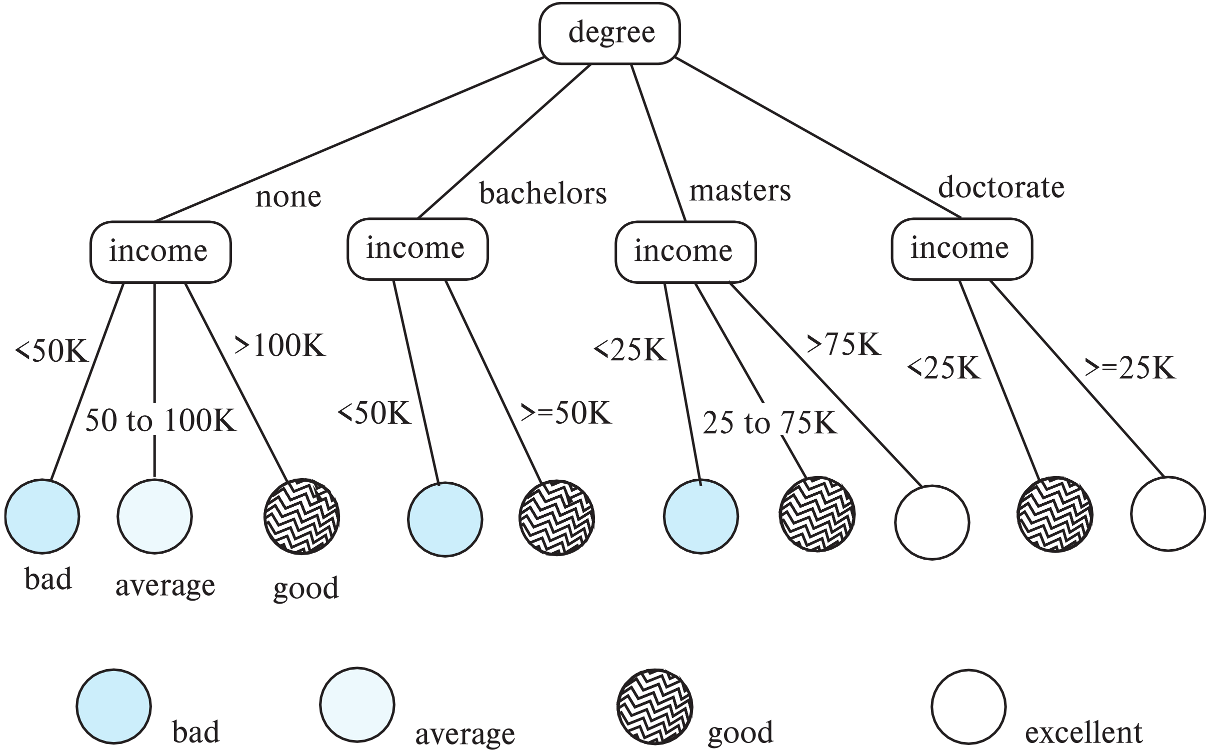
Data Mining (Cont.)

- **Descriptive Patterns**
  - **Associations**
    - Find books that are often bought by “similar” customers. If a new such customer buys one such book, suggest the others too.
    - Associations may be used as a first step in detecting **causation**
      - E.g., association between exposure to chemical X and cancer,
  - **Clusters**
    - E.g., typhoid cases were clustered in an area surrounding a contaminated well
    - Detection of clusters remains important in detecting epidemics

Classification Rules

- Classification rules help assign new objects to classes.
  - E.g., given a new automobile insurance applicant, should he or she be classified as low risk, medium risk or high risk?
- Classification rules for above example could use a variety of data, such as educational level, salary, age, etc.
  - $\forall$ person P, P.degree = masters **and** P.income > 75,000
    $\Rightarrow$ P.credit = excellent
  - $\forall$ person P, P.degree = bachelors **and**
    (P.income $\geq$ 25,000 and P.income $\leq$ 75,000)
    $\Rightarrow$ P.credit = good
- Rules are not necessarily exact: there may be some misclassifications
- Classification rules can be shown compactly as a decision tree.

Decision Tree Classifiers



Decision Trees

- Each internal node of the tree partitions the data into groups based on a **partitioning attribute**, and a **partitioning condition** for the node
- Leaf node:
  - all (or most) of the items at the node belong to the same class, or
  - all attributes have been considered, and no further partitioning is possible.
- Traverse tree from top to make a prediction
- Number of techniques for constructing decision tree classifiers
  - We omit details

Bayesian Classifiers

- Bayesian classifiers use **Bayes theorem**, which says

$$p(c_j \mid d) = \frac{p(d \mid c_j)\, p(c_j)}{p(d)}$$

where

$p(c_j \mid d)$ = probability of instance d being in class c_j,
$p(d \mid c_j)$ = probability of generating instance d given class c_j,
$p(c_j)$ = probability of occurrence of class c_j, and
$p(d)$ = probability of instance d occuring

Naïve Bayesian Classifiers

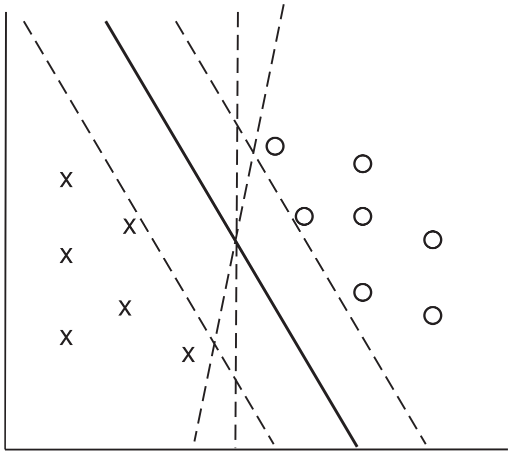
- Bayesian classifiers require
  - computation of $p\ (d \mid c_j)$
  - precomputation of $p\ (c_j)$
  - $p\ (d)$ can be ignored since it is the same for all classes
- To simplify the task, **naïve Bayesian classifiers** assume attributes have independent distributions, and thereby estimate

$$p\ (d \mid c_j) = p\ (d_1 \mid c_j) * p\ (d_2 \mid c_j) * \ldots * (p\ (d_n \mid c_j)$$

  - Each of the $p\ (d_i \mid c_j)$ can be estimated from a histogram on d_i values for each class c_j
    - the histogram is computed from the training instances
  - Histograms on multiple attributes are more expensive to compute and store

Support Vector Machine Classifiers

- Simple 2-dimensional example:
  - Points are in two classes
  - Find a line (**maximum margin line**) s.t. line divides two classes, and distance from nearest point in either class is maximum



Support Vector Machine

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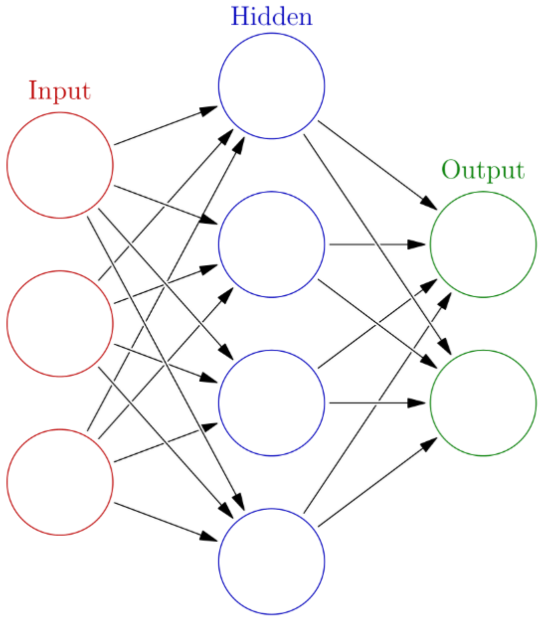
- In n-dimensions points are divided by a plane, instead of a line
- SVMs can be used separators that are curve, not necessarily linear, by transforming points before classification
  - Transformation functions may be non-linear and are called kernel functions
  - Separator is a plane in the transformed space, but maps to curve in original space
- There may not be an exact planar separator for a given set of points
  - Choose plane that best separates points
- N-ary classification can be done by N binary classifications
  - In class *i* vs. not in class *i.*

Neural Network Classifiers

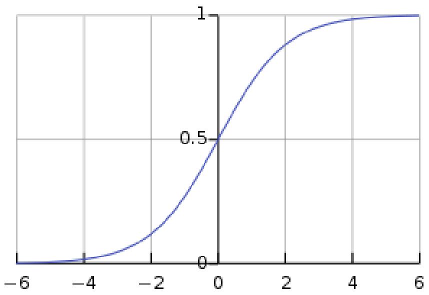
- Neural network has multiple layers
  - Each layer acts as input to next later
- First layer has input nodes, which are assigned values from input attributes
- Each node combines values of its inputs using some weight function to compute its value
  - Weights are associated with edges
- For classification, each output value indicates likelihood of the input instance belonging to that class
  - Pick class with maximum likelihood
- Weights of edges are key to classification
- Edge weights are learnt during training phase

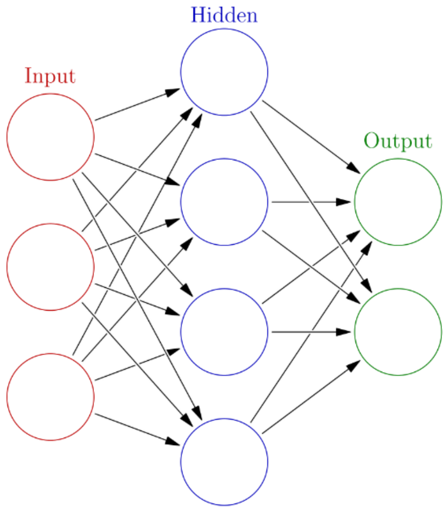


Neural Network Classifiers

- Value of a node may be linear combination of inputs, or may be a non-linear function
  - E.g., sigmoid function
- **Backpropagation algorithm** works as follows
  - Weights are set randomly initially
  - Training instances are processed one at a time
    - Output is computed using current weights
    - If classification is wrong, weights are tweaked to

      get a higher score for the correct class





Neural Networks (Cont.)

- **Deep neural networks** have a large number of layers with large number of nodes in each layer
- **Deep learning** refers to training of deep neural network on very large numbers of training instances
- Each layer may be connected to previous layers in different ways
  - Convolutional networks used for image processing
  - More complex architectures used for text processing, and machine translation, speech recognition, etc.
- Neural networks are a large area in themselves
  - Further details beyond scope of this chapter

Regression

- Regression deals with the prediction of a value, rather than a class.
  - Given values for a set of variables, $X_1, X_2, \ldots, X_n$, we wish to predict the value of a variable Y.
- One way is to infer coefficients $a_0, a_1, a_1, \ldots, a_n$ such that

$$Y = a_0 + a_1 * X_1 + a_2 * X_2 + \ldots + a_n * X_n$$

- Finding such a linear polynomial is called **linear regression**.
  - In general, the process of finding a curve that fits the data is also called **curve fitting**.
- The fit may only be approximate
  - because of noise in the data, or
  - because the relationship is not exactly a polynomial
- Regression aims to find coefficients that give the best possible fit

Association Rules

- Retail shops are often interested in associations between different items that people buy.
  - Someone who buys bread is quite likely also to buy milk
  - A person who bought the book *Database System Concepts* is quite likely also to buy the book *Operating System Concepts*.
- Associations information can be used in several ways.
  - E.g. when a customer buys a particular book, an online shop may suggest associated books.
- **Association rules**:

  bread $\Rightarrow$ *milk*　　　　*DB-Concepts, OS-Concepts* $\Rightarrow$ Networks

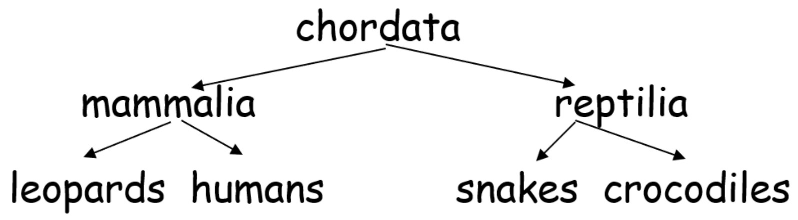
  - Left hand side: **antecedent**,　　right hand side: **consequent**
  - An association rule must have an associated **population**; the population consists of a set of **instances**
    - E.g. each transaction (sale) at a shop is an instance, and the set of all transactions is the population

Association Rules (Cont.)

- Rules have an associated support, as well as an associated confidence.
- **Support** is a measure of what fraction of the population satisfies both the antecedent and the consequent of the rule.
  - E.g., suppose only 0.001 percent of all purchases include milk and screwdrivers. The support for the rule is *milk* $\Rightarrow$ *screwdrivers* is low.
- **Confidence** is a measure of how often the consequent is true when the antecedent is true.
  - E.g., the rule *bread* $\Rightarrow$ *milk* has a confidence of 80 percent if 80 percent of the purchases that include bread also include milk.
- We omit further details, such as how to efficiently infer association rules

Clustering

- **Clustering**: Intuitively, finding clusters of points in the given data such that similar points lie in the same cluster
- Can be formalized using distance metrics in several ways
  - Group points into *k* sets (for a given *k*) such that the average distance of points from the centroid of their assigned group is minimized
    - Centroid: point defined by taking average of coordinates in each dimension.
  - Another metric: minimize average distance between every pair of points in a cluster
- **Hierarchical clustering**: example from biological classification
  - (the word classification here does not mean a prediction mechanism)



Clustering and Collaborative Filtering

- Goal: predict what movies/books/… a person may be interested in, on the basis of
  - Past preferences of the person
  - Preferences of other people
- One approach based on repeated clustering
  - Cluster people based on their preferences for movies
  - Then cluster movies on the basis of being liked by the same clusters of people
  - Again cluster people based on their preferences for (the newly created clusters of) movies
  - Repeat above till equilibrium
  - Given new user
    - Find most similar cluster of existing users and
    - Predict movies in movie clusters popular with that user cluster
- Above problem is an instance of **collaborative filtering**

Other Types of Mining

- **Text mining**: application of data mining to textual documents
- **Sentiment analysis**
  – E.g., learn to predict if a user review is positive or negative about a product
- **Information extraction**
  – Create structured information from unstructured textual description or semi-structured data such as tabular displays
- **Entity recognition** and **disambiguation**
  – E.g., given text with name “Michael Jordan” does the name refer to the famous basketball player or the famous ML expert
- **Knowledge graph** (see Section 8.4)
  – Can be constructed by information extraction from different sources, such as Wikipedia